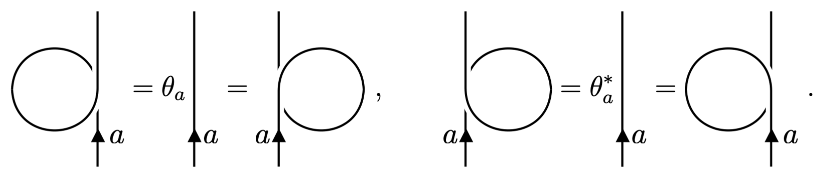
Fusion Categories and SymTFT: Homework #8

Braided Categories
March 25, 2026

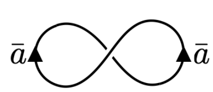
[Adapted from Andrei Bernevig & Titus Neupert's 2015 lecture notes @ IHES]

Problem 1: Topological Spin

Beyond its quantum dimension, each quasiparticle a in a braiding category is characterized by a unit-magnitude topological spin θ_a. This value represents the phase gained during a 2π self-rotation, equaling -1 for fermions or a fractional phase for other anyonic species. The diagrammatic definition of θ_a is



Alternatively, we simply connect up the open worldlines with an $\bar{a}$ worldline and obtain the definition

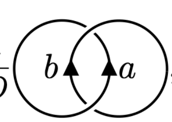
$$\theta_a := \frac{1}{d_a}$$  .

Show that this diagram can be evaluated by inserting an identity and applying an $R-$ move and get

$$\theta_a := \frac{1}{d_a} \sum_c d_c \, \mathrm{Tr} \left[R_c^{aa} \right]. \tag{1}$$

Problem 2: S-matrix

In the TQFT, the S matrix is defined diagrammatically as

$$S_{ab} := \frac{1}{D}$$  ,

where D is the total quantum dimension of the TQFT

$$D = \sqrt{\sum_a d_a^2}. \tag{2}$$

Show that, through diagrammatic manipulations, it can be written in terms of the fusion category data

$$S_{ab} = \frac{1}{D} \sum_c N_c^{ab} \operatorname{Tr}\left(R_c^{ab} R_c^{ba}\right) d_c. \tag{3}$$

1 Problem 3: Verlinde's formula (Optional)

See the discussion in Sec. III.C.2 and prove